GraphSelect: Bounded 1-Hop BFS Citation Traversal, Candidate Appraisal & Steering Topology

Hybrid Ranking: **$Score_{hybrid}(p) = 0.70 \cdot S(p) + 0.30 \cdot PR_{norm}(p)$** | Gemini 768-dim MRL Embeddings | Neo4j GDS / NetworkX

PHASE 1: BOUNDED 1-HOP BIDIRECTIONAL EXPANSION FROM SEED

🎯 Seed Paper (p_{seed})

Depth = 0 Anchor Node

- User topic / initial corpus anchor
- High citation baseline (c_p)
- Asymmetric vector embedding: e_{seed}
- OpenAlex ID / Canonical DOI

1-Hop Expansion: $N_1(p_{seed}) = N_{bwd} \cup N_{fwd}$

Backward References
N_{bwd} (cites:doi)
Prior Art, Foundations, Lineage antecedents

Co-Citation Bridge
N_{both} ($N_{bwd} \cap N_{fwd}$)
Topological bridges, Mutual grounding

Forward Citations
N_{fwd} (cited_by:doi)
Contemporary extensions, Breakthrough empiricals

Batch OpenAlex Metadata Fetch & Citation Graph Upsert into Neo4j

PHASE 2: GEMINI ASYMMETRIC EMBEDDING, PRUNING & 3-TIER CENTRALITY

3-Tier PageRank Centrality Engine

Neo4j Service Multi-Level Fallback

- **Tier 1:** Neo4j GDS (`gds.pageRank.stream`, d=0.85)
- **Tier 2:** Pure-Python NetworkX DiGraph Fallback
- **Tier 3:** In-Degree Centrality Normalization
- **Centrality Normalization:** $PR_{norm}(p) \in [0, 1]$

Asymmetric Gemini Embeddings

gemini-embedding-2 (768 dims, MRL)

- **Query:** `task: search result | query: {RQ}`
- **Doc:** `title: {title} | text: {abstract}`
- **Cosine Sim:** $S(p, q_j) = (e_p \cdot q_j) / (||e_p|| \, ||q_j||)$
- **Adaptive Pruning:** $\tau_{sim} = \mu_{sim} - \sigma_{sim}$

PHASE 3: THREE-ARCHETYPE CANDIDATE APPRAISAL ENGINE

🌉 Archetype 3: Topological Bridge
Cross-Cutting & Multi-Disciplinary Link

- **Formula:** $H_{bridge}(p) = \gamma_1 \, P\tilde{R}(p) + \gamma_2 \, \tilde{d}_p + \gamma_3 \, B_Q(p) + 0.05 \cdot \mathbb{1}(deg > 1)$
- **Weights:** $\gamma = (0.40, 0.30, 0.30)$
- **RQ Balance B_Q(p):** High similarity across ≥2 RQs
- **Profile:** Structural hub connecting distinct clusters

🏛️ Archetype 1: Foundational Prior Art
Theoretical Lineage & Seminal Grounding

- **Formula:** $H_{found}(p) = w_1 \, \tilde{c}_p + w_2 \, I(p \in N_{bwd}) + w_3 \min(1, age/5) + w_4 \, S(p) + \delta_{mature}$
- **Weights:** $w = (0.35, 0.25, 0.20, 0.20)$
- **Maturity Bonus:** $\delta_{mature} = +0.10$ if age ≥ 3 yrs
- **Profile:** Highly cited, mature, backward lineage

⚡ Archetype 2: Contemporary Breakthrough
Recent Advances & SOTA Benchmarks

- **Formula:** $H_{contemp}(p) = \mu_1 \, S_{max}(p) + \mu_2 \max(0, 1 - age/3) + \mu_3 \, \tilde{c}_p + \delta_{recent}$
- **Weights:** $\mu = (0.50, 0.35, 0.15)$
- **Velocity Bonus:** $\delta_{recent} = +0.15$ if age ≤ 2 & S ≥ 0.6
- **Profile:** Published 2024–2026, peak similarity

PHASE 4: STRATEGIC HUMAN-IN-THE-LOOP STEERING ACTIONS

Option C: Bridge Deep-Dive
Action: `expand_deeper(top_bridge)`
Uncovers inter-disciplinary connections

Option A: Foundational Deep-Dive
Action: `expand_deeper(top_foundational)`
Traces theoretical roots & seminal lineage

Option B: Contemporary Deep-Dive
Action: `expand_deeper(top_contemporary)`
Captures state-of-the-art 2025–2026 empiricals

Option D: Adopt & Conclude
Action: `add_to_library(candidates)`
Finalizes candidate pool & commits to Stage 2

🔁 Next BFS Traversal Depth (Hop = Hop + 1) — Bounded by Theorem 3.1 (|V| ≤ (b^(D+1) - 1)/(b - 1))

✅ Library Commit: Papers Persisted to Neo4j & stage_1_research.artifacts.papers (Stage 2 Analysis Ready)